

MaJo e-School Bridge

Deep Wiring & Functionality Audit Report

Generated 7/25/2026, 10:02:19 PM | Static wiring analysis + automated suite | Firebase project: majo-e-school-bridge

1. Executive verdict

Code wiring is coherent for a local-first school ERP with a cloud sync layer. Auth is designed around Cloud Function custom tokens and school claims; Firestore/Storage rules in-repo enforce role+schoolId. Automated deep suite: 18/18 passed (100%). Live multi-school selling is NOT ready until Blaze + Functions deploy + Console alignment. Dual data namespaces (AppFirestoreCollections vs DbCollections) increase wiring risk.

2. Boot & session wiring

main() !' Firebase.initializeApp !' App Check activate !' FCM background handler !' CriticalBootstrapGate !' AppBootstrap.

Critical (pre-login): registerAllDashboards, FirebaseBootstrap.tryInitialize, load Auth/Enrollment/Student/Teacher/Driver persistence, locale, school registry, login prefs, AppLock. Background: heavy LocalJsonStore loads, optional pushFullLocalStateToCloud if school claims exist, SchoolDatabaseService, FcmService, deferred Storage bootstrap.

Session restore: SessionPrefsService prefers Firebase Auth claims via SchoolAuthCloudService.restoreFromFirebaseAuth; fallback AuthService.restoreSession(username). On restore !' SessionCloudSync.startSessionWithCloudSync + NotificationService.onSessionStarted !' AuthNavigation.homeForCurrentUser().

3. Auth wiring (release vs debug)

Component	Path	Responsibility
AuthService	lib/services/auth_service.dart	Local users, session, validateLoginAsync
SchoolAuthCloudService	lib/services/school_auth_cloud_service.dart	schoolLogin callable !' custom token
FirebaseBootstrap	lib/database/firebase/firebase_bootstrap.dart	Init; claims required; anon disabled
SessionPrefsService	lib/services/session_prefs_service.dart	Persist username/school; restore
Password path	auth_secrets + callables	bcrypt server-side; client never reads secrets

Release: cloud login required when Firebase configured; fails with cloud_required if Functions unavailable. Debug: falls back to local demo accounts (kDebugMode). Forced ChangePasswordScreen when mustChangePassword. Remember-me stores identifier only (no password).

4. Sync & persistence wiring

LocalJsonStore (SharedPreferences JSON) !' FirestoreAppStore (school-scoped queries). SessionCloudSync on login pulls role-specific data then starts RealtimeMessagingBootstrap. CloudBootstrapService.ensureLoginCredentialsFromCloud is intentionally empty (no pre-login password pull). ensureAnonymousAuthReady is a misnamed alias for ensureReadyForFirestore (claims check).

Local / domain	Firestore collection
Auth profiles	app_auth_accounts
Password hashes	auth_secrets (Admin SDK only)
Students / medical	student_registry / student_medical
Teachers / drivers	teacher_registry / driver_registry
Grades / audit	grade_reports / grade_audit_log
Homework / materials	homework / learning_materials
Messages	conversations
Attendance / fees	attendance_sessions / fees
Transport live	bus_live_positions / transport_scans / ...
Inventory*	inventory_items / stock_transactions / ...
FCM	fcm_tokens
Normalized DB layer	users/classes/students/grades/... (DbCollections)

5. Deep functionality by role

Role	Modules (dashboard tiles)	Depth / notes
Teacher	classes, attendance, messages, announcements, homework, learning_materials, gallery, grades, calendar, timetable, parent, approvals, settings	Full classroom loop; grades submit for approval
Parent	children, attendance, homework, learning_materials, messages, announcements, fees, bus, grades, calendar, timetable, settings	Bus live map; fees UI partial (payments not live)
Student	profile, grades, homework, learning_materials, attendance, timetable, calendar, announcements, messages, settings	Portal can be disabled per school
Admin	add_teacher/student, approvals, portal settings, password resets, transfer, staff, students, classes, attendance, messages, announcements, finance, transport, grades, grade_approvals, workflow settings, calendar, timetable, settings	Web ! WebErpShell; workflow settings may be orphaned in grouped mobile layout
Driver	grade_approvals, workflow settings, calendar, timetable, settings	QR onboard/discharge + GPS share

6. Grades workflow wiring

Teacher Enter Grades !' SchoolDataService.submitSubjectGradeForApproval (or publish if approval off) !' Admin GradeApprovalQueueScreen approve/reject !' Parent/Student GradeReportsScreen views. Settings: AdminGradeWorkflowSettingsScreen (grade_workflow_settings). Audit: grade_audit_log.

7. Transport & messaging wiring

Driver TransportQrScannerScreen + BusLiveLocationService !' bus_live_positions. Parent bus tile !' TransportLiveMapScreen / ParentBusLinkScreen. Realtime: TransportRealtimeSync from messaging bootstrap. Messages: ConversationRealtimeSync; Cloud Function onConversationMessage fans out FCM using fcm_tokens (needs Blaze deploy). FCM client skipped on web.

8. Web ERP wiring

kIsWeb AdminDashboard !' WebErpAdminShell !' WebErpRouter.pageFor. Wired: dashboard, students, teachers, parents, employees, finance, transport, attendance, examinations/grade_approvals, grades, academic, learning_materials, announcements, calendar/events, reports, support, audit_log, system_health, settings, library, inventory, add_student/teacher. Placeholders only: institution, school, campus (+ unknown routes).

9. Firebase / cloud wiring gaps

Gap	Severity	Impact
Blaze off !' Functions undeployable	Critical	No cloud login / cross-device / FCM send
Rules may not be deployed to Console	Critical	Live DB may still be open
Anonymous still enabled in firebase.json	High	Disable in Console
App Check client on; Console not enforced	High	Bot API abuse once Functions live
Storage bootstrap deferred=true	High	Uploads stay local until billing/enable
Parent rules: school-wide student/medical read	High	Tighten to linked children
school_registry allow get: if true	Medium	School ID enumeration
Dual App vs Db collection namespaces	Medium	Sync confusion / stale paths
Misnamed ensureAnonymousAuth* APIs	Low	Maintenance hazard
grade_workflow_settings orphaned in grouped admin UI	Low	Hard to discover on mobile

10. Actual automated test report

Command: flutter test test/functionality_suite_test.dart test/database_test.dart | Exit 0 | 18/18 passed (100%)

#	Result	Test case
1	PASS	seed builds students, classes, and parent links
2	PASS	parent can reach child class teachers via resolver
3	PASS	role access rules enforce teacher and driver boundaries
4	PASS	Auth & credentials debug demo admin login succeeds locally
5	PASS	Auth & credentials wrong password rejected
6	PASS	Auth & credentials role mismatch rejected
7	PASS	Auth & credentials password policy minimum is 10
8	PASS	Auth & credentials password hash verify round-trip
9	PASS	Phone utilities normalizes Ethiopian local login key
10	PASS	Phone utilities builds E.164 for Ethiopia
11	PASS	School data / access seed builds students, classes, parent links
12	PASS	School data / access parent resolver reaches child and teachers
13	PASS	School data / access teacher class access boundaries
14	PASS	School data / access driver student access boundaries
15	PASS	Platform owner PIN PIN min length is 6
16	PASS	Platform owner PIN set and verify hashed PIN
17	PASS	Web ERP surface admin nav includes core modules
18	PASS	Web ERP surface router resolves key admin pages to widgets

11. Deep functionality health matrix

Area	Code wired	Local/debug verified	Cloud live
Role login (debug accounts)	Yes	PASS (suite)	N/A
Cloud schoolLogin	Yes	N/A (needs Functions)	BLOCKED (Blaze)
Password hashing / policy	Yes	PASS	Secrets via Functions
Parent!"child resolver	Yes	PASS	Needs claims sync
Teacher/driver access bounds	Yes	PASS	Needs claims + rules deploy
Web ERP core routes	Yes	PASS (router)	Needs hosting deploy
Web ERP org placeholders	Placeholder	N/A	N/A
Grades enter!'approve!'view	Yes	Code path	Needs cloud session
Transport QR + GPS	Yes	Code path	Needs cloud + device
Messaging + FCM fan-out	Yes	Partial	Functions blocked
Inventory / Storage uploads	Yes	Local only	Storage deferred
Online fee payments	Partial UI	Not live	Provider missing
Owner PIN console	Yes	PASS (hash/verify)	Device-local

12. Self-readiness wiring score

Architecture/wiring maturity: HIGH for offline-first ERP. Production cloud wiring: BLOCKED on Blaze + deploy. Privacy rules depth: PARTIAL (school isolation yes; relationship isolation incomplete). Product completeness: MEDIUM (core school loops yes; org placeholders + payments pending).

Recommendation: do not sell as production until (1) deploy firestore+storage rules, (2) Blaze+Functions+admin bootstrap, (3) parent/medical rule tightening, (4) multi-device pilot pass. Safe to sell only as supervised pilot with local/debug caveats.

13. Recommended next wiring fixes

1. Rename ensureAnonymousAuth* !' ensureSchoolAuth* across call sites.
2. Deploy rules; disable Anonymous in Console; flip firebase.json.
3. Blaze !' deploy functions !' seed admin !' migrateAuthSecrets.
4. Tighten Firestore parent reads to linkedStudentIds.
5. Enable Storage bootstrap when bucket is live; deploy storage.rules.
6. Surface grade_workflow_settings in admin grouped sections.
7. Document/clarify AppFirestoreCollections vs DbCollections ownership.
8. Re-run this PDF generator after each deploy milestone.

JSON: [reports/wiring_functionality_test_results.json](#)

PDF: [MaJo_Wiring_Functionality_Audit.pdf](#)